BIOMETRY – AAA 200

FREQUENCY DISTRIBUTION

DEFINITION

A frequency distribution is a way to organize and display data to show how often each value or range of values occurs.

There are several ways to represent frequency distributions, including:

1. **Tables**: A frequency table lists each value or range of values and the number of times it occurs.
2. **Histograms**: A histogram is a type of bar chart that represents the frequency of values within specified ranges or intervals. The height of each bar represents the frequency of values within that interval.
3. **Pie charts**: A pie chart can display the relative frequency of different categories as slices of a whole pie.
4. **Frequency polygons**: This is a line graph that shows the frequency of each value or interval, typically by connecting the midpoints of each interval with straight lines.

Frequency distributions are useful in statistics because they help to visualize and understand the distribution of data, identify patterns, and detect any outliers or anomalies.

A frequency distribution is a summary of data that shows the number of observations (or frequency) of each value or range of values in a dataset. It organizes data into classes or intervals and counts how many data points fall into each class. Frequency distributions are often presented in tables or graphs (e.g., histograms, bar charts) to make the data easier to understand and analyze. It is essentially a summary of the frequency of individual values or ranges of values in a dataset.

Key Components of a Frequency Distribution:

1. **Classes (or bins)**: Ranges of values into which the data is grouped.
2. **Frequency**: The number of data points that fall into each class.
3. **Class width**: The size of each class interval.

4. **Cumulative frequency**: The running total of frequencies up to each class.
5. **Relative frequency**: The proportion of data points in each class (frequency divided by total number of data points).

Steps to Create a Frequency Distribution:

1. **Determine the range**: Find the minimum and maximum values in the dataset. Subtract the minimum from the maximum.
2. **Choose the number of classes**: Decide how many classes (or bins) to use.
3. **Calculate class width**: Divide the range by the number of classes.
4. **Create class intervals**: Define the ranges for each class.
5. **Count frequencies**: Count how many data points fall into each class.
6. **Optional**: Calculate cumulative frequency and relative frequency.

Example of a Frequency Distribution:
Suppose you have the following dataset representing the ages of 20 people:

23,25,28,29,30,31,32,34,35,36,37,38,40,42,43,45,47,50,52,55

Step 1: Determine the Range

- Minimum value: 23
- Maximum value: 55
- Range $55-23=32$

Step 2: Choose the Number of Classes

- Let's use 5 classes.

Step 3: Calculate Class Width

- Class width:$32/5=6.4$ $32 = 6.4$ (round up to 7 for simplicity).

Step 4: Create Class Intervals

- 20-26, 27-33, 34-40, 41-47, 48-55.

Step 5: Count Frequencies

Class Interval	Frequency
20-26	2
27-33	5
34-40	6
41-47	4
48-55	3

Step 6: Optional Calculations

- **Cumulative frequency**: Running total of frequencies.
- **Relative frequency**: Frequency divided by total number of data points (20).
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Class Interval	Frequency	Cumulative Frequency	Relative Frequency
20-26	2	2	0.10
27-33	5	7	0.25
34-40	6	13	0.30

Class Interval	Frequency	Cumulative Frequency	Relative Frequency
41-47	4	17	0.20
48-55	3	20	0.15

Example 2: Frequency Distribution for Exam Scores

Suppose you have the following exam scores for 15 students:
55,60,65,70,72,75,78,80,82,85,88,90,92,95,100

Step 1: Determine the Range
- Minimum value: 55
- Maximum value: 100
- Range: $100-55=45$

Step 2: Choose the Number of Classes
- Let's use 5 classes.

Step 3: Calculate Class Width
- Class width: $45/5=9$.

Step 4: Create Class Intervals
- 55-63, 64-72, 73-81, 82-90, 91-100.

Step 5: Count Frequencies

Class Interval	Frequency
55-63	2
64-72	4
73-81	5
82-90	6
91-100	3

Step 6: Optional Calculations
- **Cumulative frequency**: Running total of frequencies.
- **Relative frequency**: Frequency divided by total number of data points (15).

Class Interval	Frequency	Cumulative Frequency	Relative Frequency
55-63	2	2	0.13
64-72	4	6	0.27
73-81	5	11	0.33
82-90	6	17	0.40
91-100	3	20	0.20

Uses of Frequency Distributions:
1. **Summarize data**: Organize large datasets into manageable groups.
2. **Identify patterns**: Reveal trends, clusters, or gaps in the data.
3. **Compare datasets**: Compare distributions of different datasets.
4. **Prepare for visualization**: Serve as the basis for histograms, bar charts, or other graphs.

Example 3: Frequency Distribution for Categorical Data
Suppose you have the following data representing the favorite colors of 20 people:

1. 1 Red
2. 2 Blue
3. 3 Green
4. 4 Blue
5. 5 Red
6. 6 Yellow
7. Green
8. Blue
9. Red
10. Green
11. Blue
12. Yellow Red
13. Blue
14. Green
15. Red
16. Blue
17. Green
18. Yellow
19. Red
20. Red

Step 1: Create Categories

- Categories: Red, Blue, Green, Yellow.

Step 2: Count Frequencies

Color	Frequency
Red	
Blue	
Green	
Yellow	

Step 3: Optional Calculations

- **Relative frequency**: Frequency divided by total number of data points (20).

Color	Frequency	Relative Frequency
Red		
Blue		
Green		
Yellow		

HISTOGRAMS

A **histogram** is a graph showing the distribution of numerical data. It estimates the probability distribution of a continuous variable by dividing data into intervals (**bins**) and counting how many data points fall into each bin. These bins are shown as adjacent rectangles, with heights representing the frequency of data points in each bin.

Key Features of a Histogram

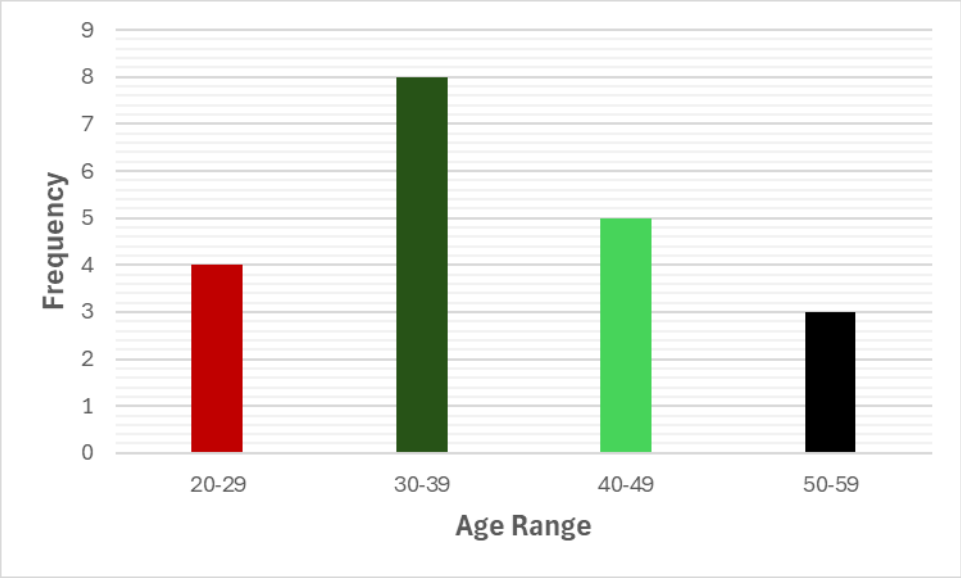
a. **Bins**: The range of values divided into intervals.
b. **Frequency**: The number of data points in each bin.
c. **Bars**: Rectangles representing the frequency of each bin.
d. **X-axis**: Represents the range of the data (bins).
e. **Y-axis**: Represents the frequency or count of data points in each bin.

Example of a Histogram:
Suppose you have the following dataset representing the ages of 20 people:
23, 25, 28, 29, 30, 31, 32, 34, 35, 36, 37, 38, 40, 42, 43, 45, 47, 50, 52, 55

Steps to Create a Histogram:

1. **Choose bin ranges**: For example, bins of size 10: 20-29, 30-39, 40-49, 50-59.
2. **Count frequencies**:
   - 20-29: 4 people (23, 25, 28, 29)
   - 30-39: 8 people (30, 31, 32, 34, 35, 36, 37, 38)
   - 40-49: 5 people (40, 42, 43, 45, 47)
   - 50-59: 3 people (50, 52, 55)
3. **Plot the histogram**:
   - X-axis: Age ranges (20-29, 30-39, 40-49, 50-59)
   - Y-axis: Frequency (4, 8, 5, 3)

The histogram will show taller bars for the 30-39 age group (highest frequency) and shorter bars for the 50-59 age group (lowest frequency).



Uses of Histograms

1. **Visualize data distribution**: Helps identify patterns, skewness, or outliers.
2. **Compare datasets**: Compare distributions of different datasets.
3. **Identify central tendency**: Shows where most data points lie.
4. **Understand spread**: Reveals the variability in the data.

Histogram vs Bar Chart:

- **Histogram**: Used for continuous data, bars touch each other.
- **Bar Chart**: Used for categorical data, bars are separated.

Histograms are a powerful tool in statistics for understanding the shape, center, and spread of data!

FREQUENCY POLYGON

A frequency polygon is a graphical representation of a dataset's distribution, similar to a histogram. It is created by plotting the frequencies of various classes (or bins) of data and connecting these points with straight lines. Frequency polygons are useful for visualizing the shape of the distribution and for comparing multiple datasets.

Key Features of a Frequency Polygon:

a. **Points**: Each point represents the frequency of a class (bin) at its midpoint.
b. **Lines**: The points are connected by straight lines.
c. **X-axis**: Represents the midpoints of the classes (bins).
d. **Y-axis**: Represents the frequency (count) of data points in each class.
e. **Closed Shape**: The polygon is often closed by connecting the ends to the X-axis (zero frequency) for better visualization.

Steps to Create a Frequency Polygon:

a. Create a frequency distribution table: Divide the data into classes (bins) and count the frequency of each class.
b. Find the midpoints of each class: Calculate the midpoint of each bin.
c. Plot the points: On the X-axis, use the midpoints of the classes, and on the Y-axis, use the frequencies.
d. Connect the points: Use straight lines to connect the points in order.
e. Close the polygon (optional): Extend the lines to the X-axis at both ends to form a closed shape.

Example of a Frequency Polygon

Consider the following dataset representing the ages of 20 people:

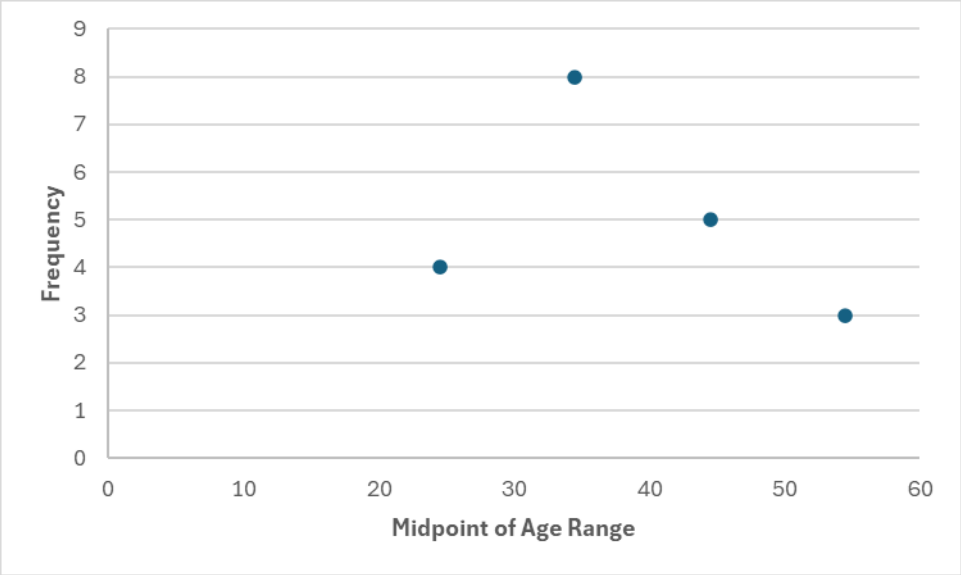
23, 25, 28, 29, 30, 31, 32, 34, 35, 36, 37, 38, 40, 42, 43, 45, 47, 50, 52, 55

Step 1: Create a Frequency Distribution Table

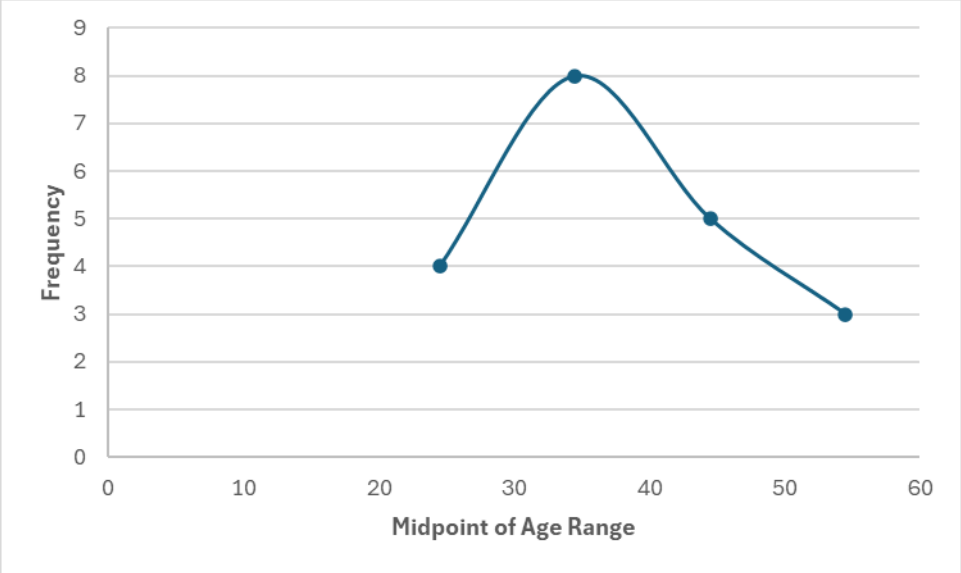
Class (Age Range)	Midpoint	Frequency
20-29	24.5	4
30-39	34.5	8
40-49	44.5	5
50-59	54.5	3

Step 2: Plot the Points

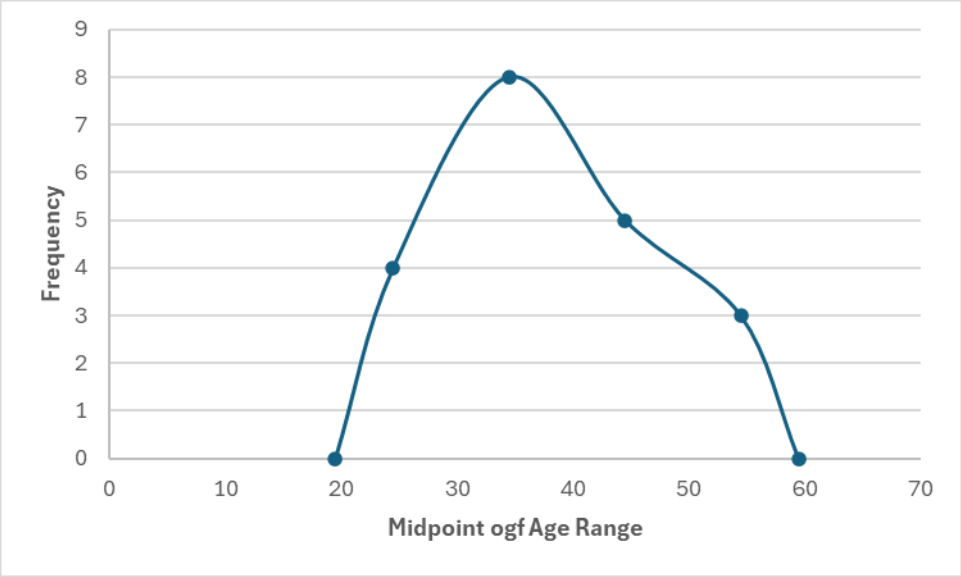
a. Plot the midpoints (24.5, 34.5, 44.5, 54.5) on the X-axis.
b. Plot the frequencies (4, 8, 5, 3) on the Y-axis.



Step 3: Connect the Points
Connect the points (24.5, 4), (34.5, 8), (44.5, 5), and (54.5, 3) with straight lines.



Step 4: Close the Polygon (Optional)
Extend the lines to the X-axis at the start (19.5, 0) and end (59.5, 0) to form a closed shape.



Uses of Frequency Polygons:

a. **Compare datasets**: Overlay multiple frequency polygons to compare distributions.
b. **Visualize trends**: Easily identify peaks, valleys, and patterns in the data.
c. **Smooth representation**: Provides a smoother visualization than histograms.

Frequency Polygon vs Histogram:

a. **Frequency Polygon**: Uses lines to connect points, better for comparing multiple datasets.
b. **Histogram**: Uses bars, better for visualizing the distribution of a single dataset.

Frequency polygons are a great tool for understanding the shape and trends of data distributions!

OGIVES

An **ogive**, also known as a **cumulative frequency curve or line graph**, is a graphical representation of the cumulative frequencies for a dataset. It displays the running total of frequencies up to each class (or bin) in a frequency distribution. Ogives are used to understand the cumulative distribution of data and to estimate percentiles, medians, and quartiles.

Key Features of an Ogive:

a. **Cumulative Frequency**: The running total of frequencies up to each class.
b. **Points**: Each point represents the cumulative frequency at the upper boundary of a class.
c. **Lines**: The points are connected by straight lines to form a curve.
d. **X-axis**: Represents the upper boundaries of the classes (bins).
e. **Y-axis**: Represents the cumulative frequency.

Types of Ogives:

a. Less Than Ogive: Shows the cumulative frequency less than or equal to the upper boundary of each class.
b. Greater Than Ogive: Shows the cumulative frequency greater than or equal to the lower boundary of each class.

Steps to Create an Ogive:

a. Create a frequency distribution table: Divide the data into classes (bins) and count the frequency of each class.
b. Calculate cumulative frequencies: Add the frequencies sequentially to get the running total.
c. Plot the points: On the X-axis, use the upper boundaries of the classes, and on the Y-axis, use the cumulative frequencies.
d. Connect the points: Use straight lines to connect the points in order.

Example of an Ogive:

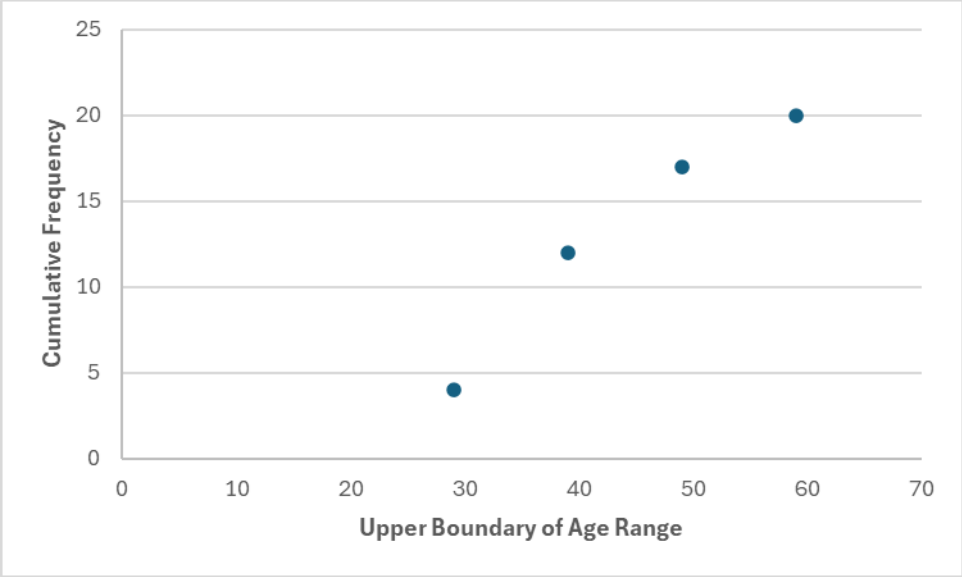
Consider the following dataset representing the ages of 20 people:
23,25,28,29,30,31,32,34,35,36,37,38,40,42,43,45,47,50,52,55

Step 1: Create a Frequency Distribution Table

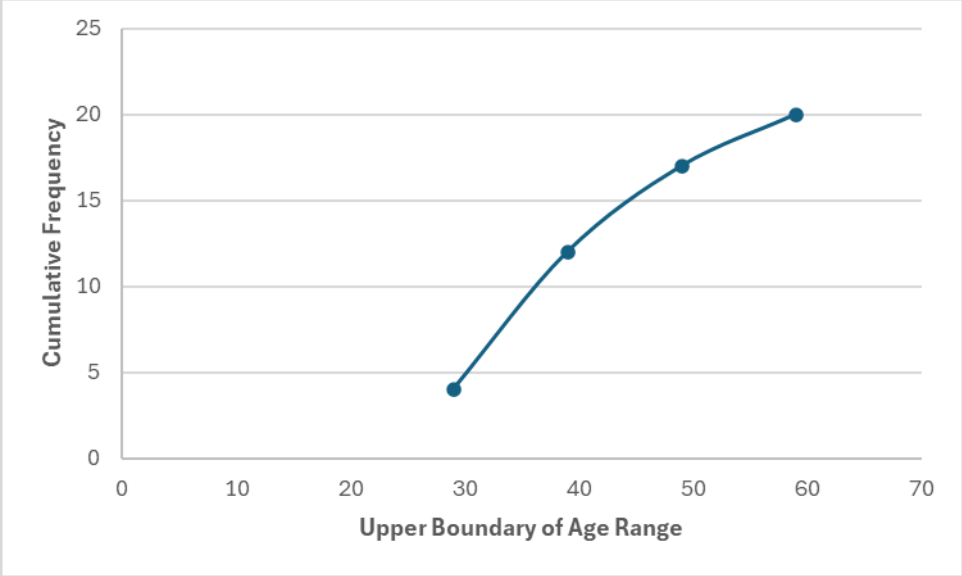
Class (Age Range)	Upper Boundary	Frequency	Cumulative Frequency
20-29	29	4	4
30-39	39	8	12 (4 + 8)
40-49	49	5	17 (12 + 5)
50-59	59	3	20 (17 + 3)

Step 2: Plot the Points

a. Plot the upper boundaries (29, 39, 49, 59) on the X-axis.
b. Plot the cumulative frequencies (4, 12, 17, 20) on the Y-axis.



Step 3: Connect the Points
Connect the points (29, 4), (39, 12), (49, 17), and (59, 20) with straight lines.



Uses of an Ogive:

a. Estimate percentiles: Determine the value below which a given percentage of data falls.
b. Find the median: The point where the cumulative frequency is half the total frequency.
c. Compare distributions: Compare cumulative distributions of different datasets.
d. Understand data spread: Visualize how data accumulates over the range.

Ogive vs Frequency Polygon:

a. Ogive: Represents cumulative frequencies, used for percentiles and medians.
b. Frequency Polygon: Represents individual frequencies, used for visualizing data distribution.

Ogive curves are a powerful tool for analyzing cumulative data and understanding the overall distribution of a dataset!

DOT PLOTS

A **dot plot** (or dot chart) is a graphical method for representing data, where individual data points are marked by dots along an axis. Dot plots are particularly effective for illustrating the distribution of small to moderate-sized datasets. They serve as a valuable tool for identifying patterns, clusters, gaps, and outliers within the data.

Key Features of a Dot Plot:

a. Axis: A single axis (usually horizontal) represents the range of the data.
b. Dots: Each dot represents one or more data points.
c. Stacked Dots: If multiple data points have the same value, the dots are stacked vertically.
d. Simplicity: Easy to create and interpret.

Steps to Create a Dot Plot:

a. Draw a horizontal axis: Label it with the range of the data.
b. Plot the dots: For each data point, place a dot above the corresponding value on the axis.
c. Stack dots: If multiple data points have the same value, stack the dots vertically.

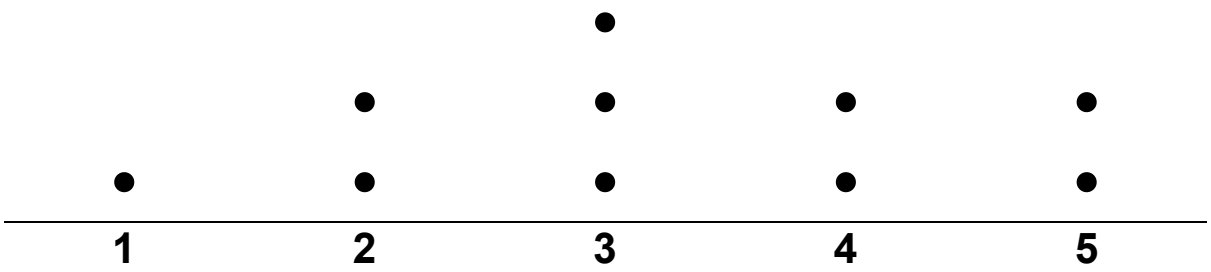
Example of a Dot Plot:
Consider the following dataset representing the number of goals scored by a soccer team in 10 matches:
1,2,2,3,3,3,4,4,5,5

Step 1: Draw the Axis
Draw a horizontal axis labeled from 1 to 5 (the range of the data).

Step 2: Plot the Dots
For each value, place a dot above the corresponding number on the axis:

a. 1: One dot
b. 2: Two dots
c. 3: Three dots
d. 4: Two dots
e. 5: Two dots

The dot plot will look like this:



Uses of Dot Plots:

a. Visualize small datasets: Ideal for datasets with fewer than 50 data points.
b. Identify patterns: Easily spot clusters, gaps, and outliers.
c. Compare groups: Overlay multiple dot plots to compare distributions.

d. Simple and clear: Easy to create and interpret, even for non-technical audiences.

Dot Plot vs Histogram:

a. Dot Plot: Represents individual data points, best for small datasets.
b. Histogram: Groups data into bins, best for larger datasets.

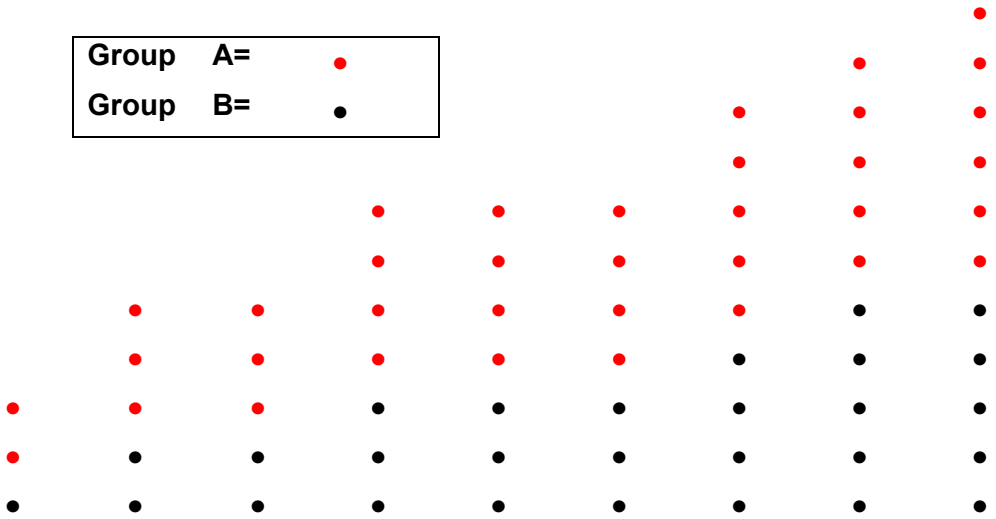
Example 3: Dot Plot for Comparing Groups

Suppose you have the following data for the number of hours studied by two groups of students:

Group A: 2, 3, 3, 4, 4, 4, 5, 5, 6
Group B: 1, 2, 2, 3, 3, 4, 4, 5, 5

Create a Dot Plot for Each Group:

a. Use different colors or symbols for each group.
b. Plot the dots for Group A and Group B on the same axis.



The dot plot will show the distribution of study hours for both groups, making it easy to compare Group A students' study hours against Group B students.

Dot plots are a simple yet powerful tool for visualizing and analyzing data!

STEM AND LEAF PLOTS

A **stem-and-leaf plot** (or stemplot) is a graph that displays numerical data, showing both the distribution and individual data points. It is useful for small to moderate datasets and quickly visualizes the data's shape.

Key Features of a Stem-and-Leaf Plot:

a. **Stems**: The leading digit(s) of the data points (e.g., the tens place).
b. **Leaves**: The trailing digit(s) of the data points (e.g., the ones place).
c. **Ordered Display**: Data is sorted in ascending order within each stem.
d. **Simplicity**: Easy to construct and interpret.

Steps to Create a Stem-and-Leaf Plot:

a. Separate each data point into a stem (leading digit(s)) and a leaf (trailing digit(s)).
b. List the stems in a vertical column from smallest to largest.
c. Add the leaves to the right of each stem, ordered from smallest to largest.
d. Include a key to explain the interpretation of the stems and leaves.

Example of a Stem-and-Leaf Plot:

Consider the following dataset representing the ages of 15 people:

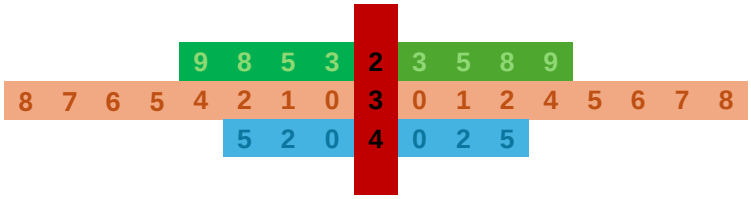
23, 25, 28, 29, 30, 31, 32, 34, 35, 36, 37, 38, 40, 42, 45

Step 1: Separate Stems and Leaves

a. Stems: Tens digits (2, 3, 4) i.e. there 20s, 30s and 40s
b. Leaves: Ones digits (0-9) i.e. there is 0 (on 30 & 40) and a 9 (on 29) making the range to be lowest=0 and highest=9.

Step 2: Create the Plot

Tens	**Ones (i.e. tens 2<u>3</u> and ones 2<u>3</u>)**							
2	**3**	**5**	**8**	**9**				
3	**0**	**1**	**2**	**4**	**5**	**6**	**7**	**8**
4	**0**	**2**	**5**					



Step 3: Add a Key

- Key: 2 | 3 means 23.

Uses of Stem-and-Leaf Plots:

a. Visualize data distribution: Shows the shape, spread, and outliers.
b. Preserve raw data: Each data point is visible in the plot.

c. Compare datasets: Side-by-side stem-and-leaf plots can compare two datasets.
d. Quick analysis: Easy to construct and interpret.

Example 3: Stem-and-Leaf Plot with Decimals

Suppose you have the following dataset representing weights (in kg) of 10 objects:

5.3, 5.7, 6.1, 6.2, 6.4, 6.8, 7.0, 7.2, 7.5, 7.9

Step 1: Separate Stems and Leaves
a. Stems: Units digit (5, 6, 7).
b. Leaves: Tenths digit (0-9).

Step 2: Create the Plot

5 | 3 7
6 | 1 2 4 8
7 | 0 2 5 9

Step 3: Add a Key
- Key: 5 | 3 means 5.3.

Example 4: Back-to-Back Stem-and-Leaf Plot (Comparing Two Groups)

Suppose you have the following data for the number of hours studied by two groups of students:

Group A: 2, 3, 3, 4, 4, 4, 5, 5, 6
Group B: 1, 2, 2, 3, 3, 4, 4, 5, 5

Step 1: Create the Plot

Group A	Stem	Group B
	1	
	2	1
3 3	3	2 2 3 3
4 4 4	4	4 4
5 5	5	5 5
6	6	

Step 2: Add a Key
Key: 4 | 5 means 45.

Advantages of Stem-and-Leaf Plots:
a. Retains raw data: Each data point is visible.
b. Easy to construct: No complex calculations required.
c. Quick insights: Provides a clear picture of the data distribution.

Stem-and-leaf plots are a simple yet effective way to organize and visualize data!

PIE CHARTS

A **pie chart** is a circular graphic divided into slices, each representing a category's proportion in a dataset. The size of each slice reflects the quantity or percentage it represents. Pie charts are effective for displaying data composition, especially when there are fewer than six categories.

Key Features of a Pie Chart:

a. **Circle**: Represents the whole dataset (100%).
b. **Slices**: Each slice represents a category and its proportion of the whole.
c. **Labels**: Categories and their percentages or values are often labeled for clarity.
d. **Colours**: Different colours or patterns are used to distinguish between slices.

Steps to Create a Pie Chart:

a. **Collect data:** Gather the data for each category.
b. **Calculate proportions:** Determine the percentage or fraction of the whole that each category represents.
c. **Draw the circle:** Use a compass or software to draw a circle.
d. **Divide the circle:** Divide the circle into slices based on the proportions of each category.
e. **Label the slices:** Add labels, percentages, or values to each slice.
f. **Add a legend:** Include a legend to explain the colors or patterns used for each category.

Example of a Pie Chart:
Suppose you have the following data representing the favorite fruits of 40 people:

Fruit	Number of People	Percentage
Apple	10	25%
Banana	8	20%
Orange	12	30%
Mango	6	15%
Grapes	4	10%

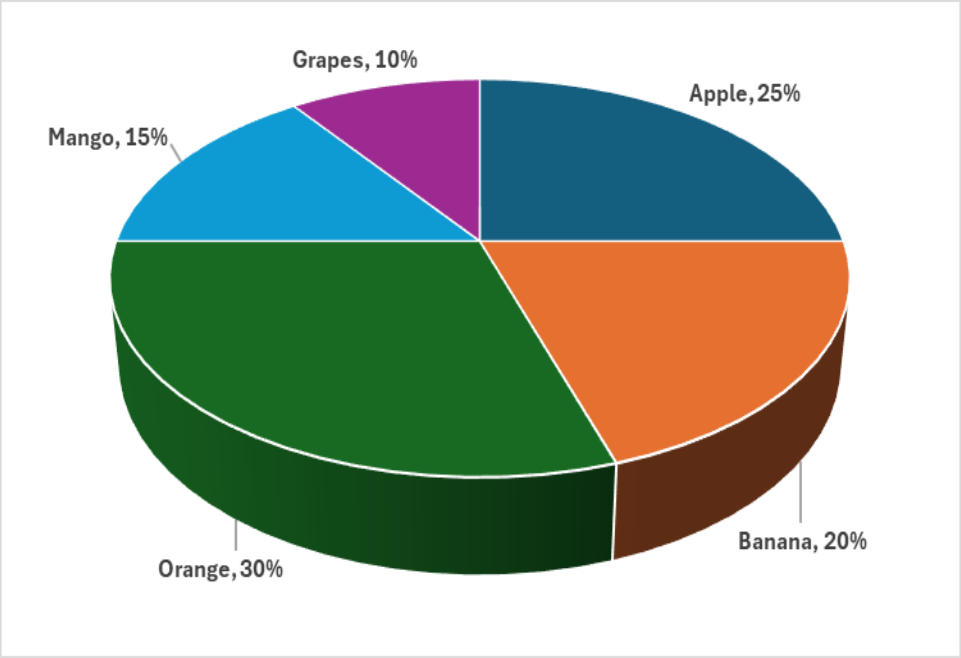
Step 1: Calculate Percentages

a. Apple: 10/40×100 = 25%
b. Banana: 8/40×100 = 20%
c. Orange: 12/40×100 = 30%
d. Mango: 6/40×100 = 15%
e. Grapes: 4/40×100 = 10%

Step 2: Draw the Pie Chart

- Draw a circle and divide it into slices based on the percentages:
  Apple: 25% (90°)
  Banana: 20% (72°)
  Orange: 30% (108°)
  Mango: 15% (54°)
  Grapes: 10% (36°)

Step 3: Label the Slices
Add labels and percentages to each slice.



Uses of Pie Charts:

a. Show proportions: Visualize the composition of a whole.
b. Compare categories: Highlight the relative sizes of different categories.
c. Simple representation: Easy to understand for non-technical audiences.

Limitations of Pie Charts:

a. Too many categories: Becomes cluttered and hard to read if there are more than 6 categories.
b. Small differences: Hard to distinguish between slices with similar proportions.
c. No exact values: Percentages or labels are needed to interpret the data accurately.

Pie charts are a simple and effective way to visualize proportions and make data easy to understand!

BAR CHARTS

A **bar chart** (or bar graph) is a graphical representation of data utilizing rectangular bars. The length or height of each bar corresponds to the value it represents. Bar charts are employed to compare different categories or groups of data. They can be oriented vertically (column chart) or horizontally, depending on context and preference.

Key Features of a Bar Chart:

a. **Bars**: Rectangles represent the values of each category.
b. **Axes**:
   - **X-axis**: Represents the categories or groups.
   - **Y-axis**: Represents the values or frequencies.
c. **Labels**: Categories and values are labeled for clarity.
d. **Colors**: Different colors or patterns are used to distinguish between bars (if multiple datasets are shown).

Types of Bar Charts:

a. Vertical Bar Chart: Bars are displayed vertically (also called a column chart).
b. Horizontal Bar Chart: Bars are displayed horizontally.
c. Grouped Bar Chart: Multiple datasets are grouped together for comparison.
d. Stacked Bar Chart: Bars are divided into sub-parts to show the composition of each category.

Steps to Create a Bar Chart:

a. **Collect data**: Gather the data for each category.
b. **Choose orientation**: Decide whether to use a vertical or horizontal bar chart.
c. **Draw axes**: Label the X-axis (categories) and Y-axis (values).
d. **Draw bars**: For each category, draw a bar with a height or length proportional to its value.
e. **Add labels**: Label the bars with their corresponding values or percentages.
f. **Add a legend**: If multiple datasets are shown, include a legend to explain the colors or patterns.

Example of a Bar Chart:

Suppose you have the following data representing the sales of different products in a store:

Product	Sales (in K)
A	200
B	300
C	150
D	400

Step 1: Draw the Axes

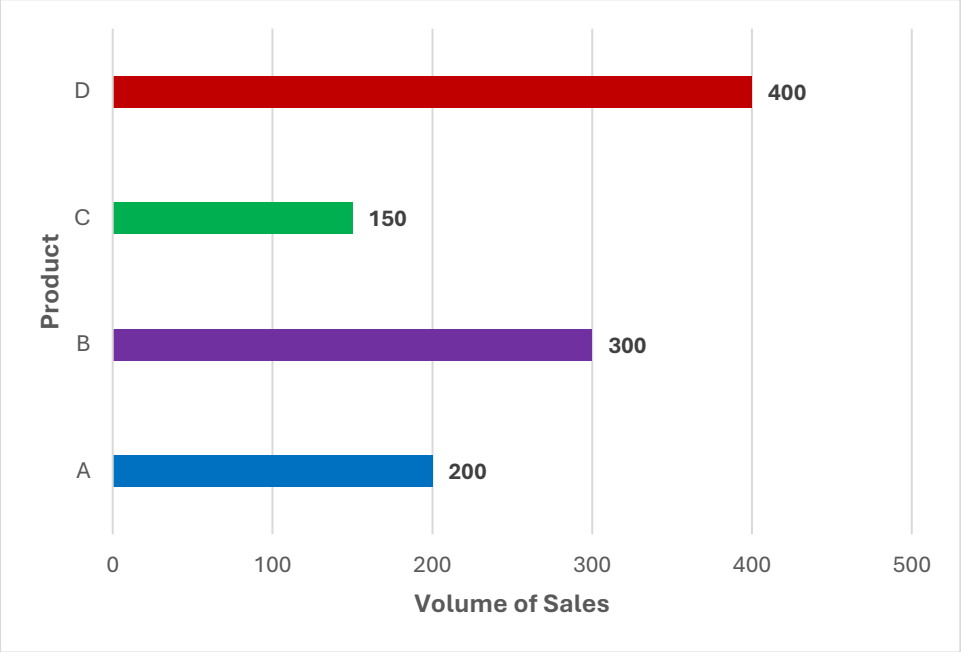
a. X-axis: Products (A, B, C, D).
b. Y-axis: Sales (in dollars).

Step 2: Draw the Bars

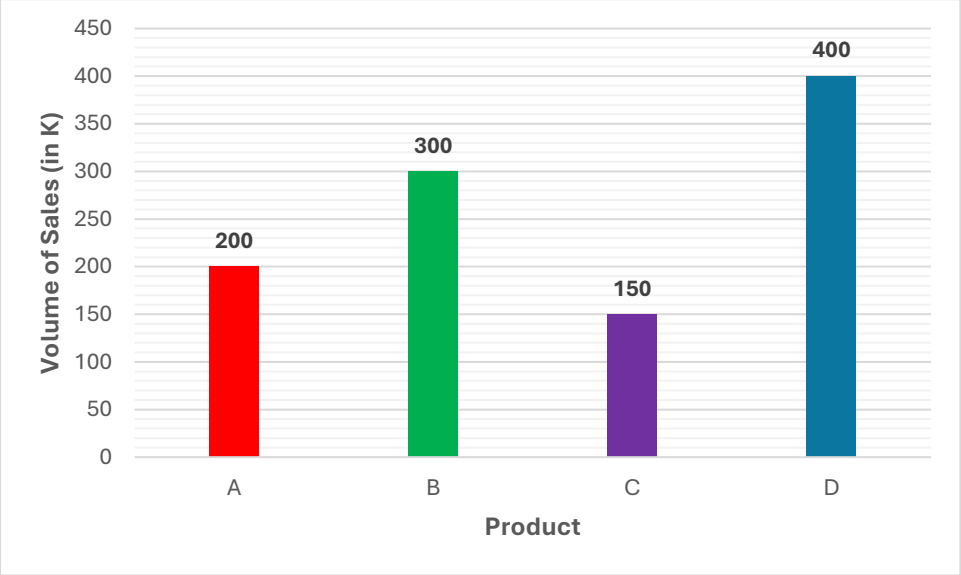
a. For Product A: Draw a bar up to 200 on the Y-axis.
b. For Product B: Draw a bar up to 300 on the Y-axis.

c. For Product C: Draw a bar up to 150 on the Y-axis.
d. For Product D: Draw a bar up to 400 on the Y-axis.

Horizontal Bar Chart



Vertical Bar Chart



Step 3: Add Labels

- Label each bar with its corresponding sales value.

Example 2: Grouped Bar Chart

Suppose you have the following data representing the sales of two products (X and Y) over three months:

Month	Product X Sales (in K)	Product Y Sales (in K)
Jan	100	150
Feb	200	250
Mar	300	350

Step 1: Draw the Axes

a. X-axis: Months (Jan, Feb, Mar).
b. Y-axis: Sales (in dollars).

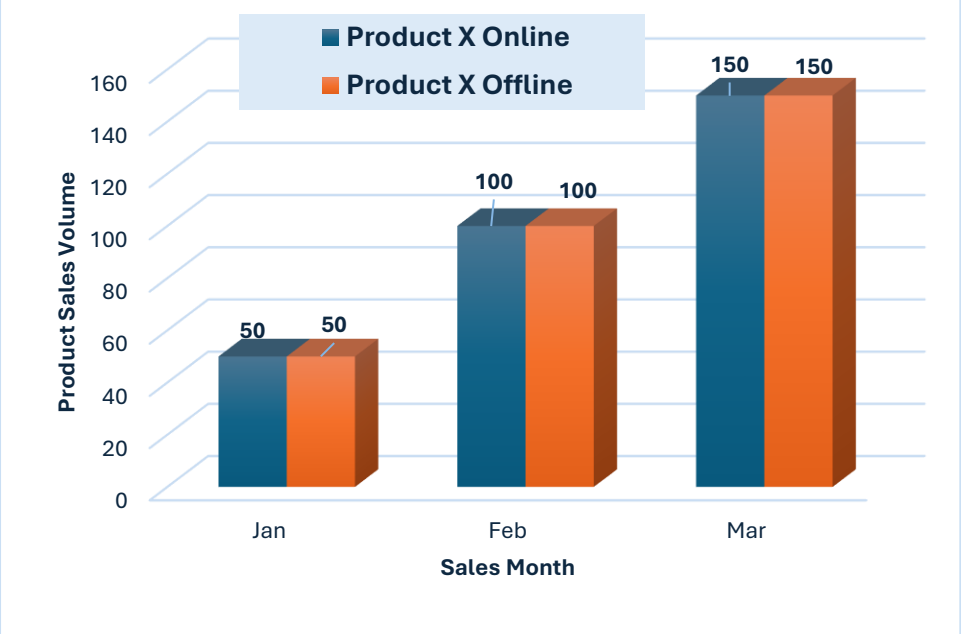
Step 2: Draw the Bars

For each month, draw two bars side by side (one for Product X and one for Product Y):

a. Jan: Product X (100), Product Y (150).
b. Feb: Product X (200), Product Y (250).
c. Mar: Product X (300), Product Y (350).

Step 3: Add Labels and Legend

a. Label the bars with their corresponding sales values.
b. Add a legend to distinguish between Product X and Product Y.



Example 3: Stacked Bar Chart

Suppose you have the following data representing the sales of two products (X and Y) over three months, with a breakdown of online and offline sales:

Month	Product X Online	Product X Offline	Product Y Online	Product Y Offline
Jan	50	50	75	75
Feb	100	100	125	125
Mar	150	150	175	175

Step 1: Draw the Axes

a. X-axis: Months (Jan, Feb, Mar).
b. Y-axis: Sales (in dollars).

Step 2: Draw the Bars

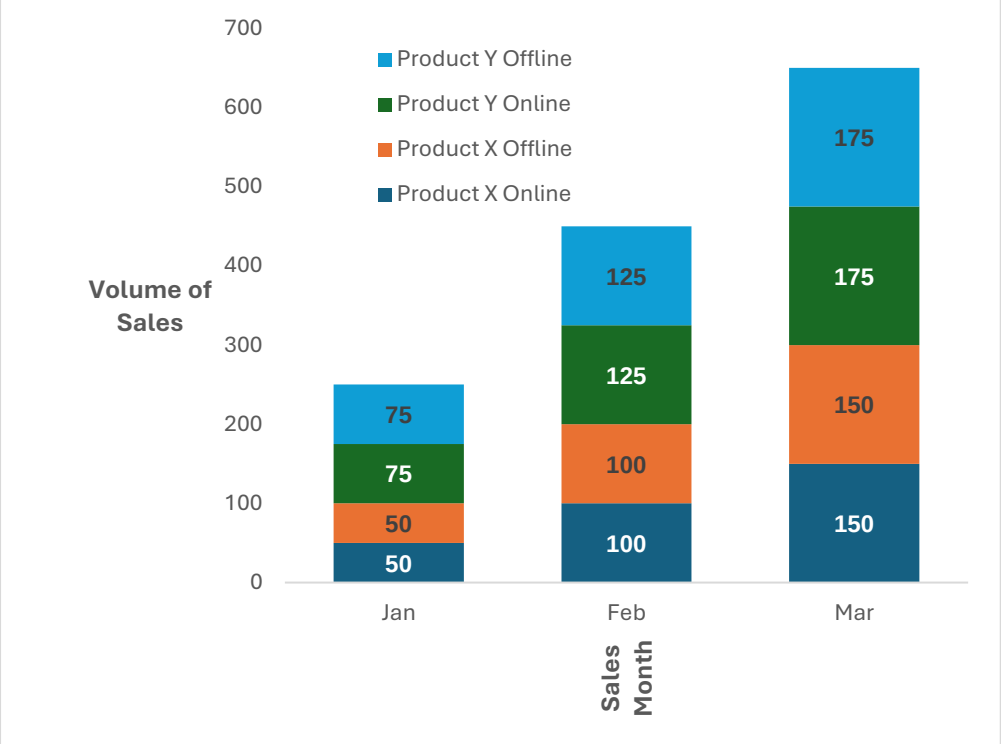
For each month, draw a stacked bar:

a. Jan: Product X (50 + 50), Product Y (75 + 75).
b. Feb: Product X (100 + 100), Product Y (125 + 125).
c. Mar: Product X (150 + 150), Product Y (175 + 175).

Step 3: Add Labels and Legend

a. Label the bars with their corresponding sales values.
b. Add a legend to distinguish between online and offline sales.

Month	Product X Online	Product X Offline	Product Y Online	Product Y Offline
Jan	50	50	75	75
Feb	100	100	125	125
Mar	150	150	175	175



Uses of Bar Charts:

a. **Compare categories**: Easily compare values across different categories.
b. **Show trends**: Visualize trends over time or across groups.
c. **Highlight differences**: Emphasize differences between groups or categories.
d. **Simple and clear**: Easy to create and interpret.